1 Architecture

1.1 Atlas Autoencoder

Definition 1.1 (Atlas) *Let x be a data point on a d-dimensional manifold $M \subset \mathbb{R}^D$. A chart C_i is a diffeomorphism $C_i : \mathbb{R}^d \longrightarrow M$. The finite family $\{C_i\}_{i=1}^m$ is the* atlas.

Our block is a *mixture-of-autoencoders*. We learn

1. A shared gating network $g : \mathbb{R}^D \longrightarrow \mathbb{R}^m$ producing logits $z_i = g_i(x)$. Soft-max yields soft chart weights $\alpha_i = \mathrm{softmax}(z)_i$.
2. For every chart i a pair of MLPs

$$C_i : \mathbb{R}^d \longrightarrow M, \qquad C_i(\xi) = \sigma\big(W_{C_i}\xi + b_{C_i}\big),$$

$$D_i : M \longrightarrow \mathbb{R}^d, \qquad D_i(x) = \sigma\big(W_{D_i}x + b_{D_i}\big),$$

   with $C_i^{-1} \approx D_i$ and σ the ReLU.

For each chart we reconstruct

$$\xi_i = D_i(x),$$
$$\hat{x}_i = C_i(\xi_i),$$

then recombine

$$\hat{x} \;=\; \sum_{i=1}^{m} \alpha_i\, \hat{x}_i.$$

We minimize an autoencoder reconstruction loss and a per-sample entropy that mitigates dead charts

$$\mathcal{L}_{\mathrm{rec}} = \|\hat{x} - x\|_2^2, \qquad \mathcal{L}_{\mathrm{ent}} = -\sum_{i=1}^{m} \alpha_i \log \alpha_i, \qquad \mathcal{L} = \mathcal{L}_{\mathrm{rec}} + \lambda\, \mathcal{L}_{\mathrm{ent}}.$$

1.2 Betti Numbers

Definition 1.2 (Overlap weights) *For charts i, j define*

$$w_{ij} \;=\; \mathbb{E}_{x\sim\mathcal{D}}\big[\min\big(\alpha_i(x), \alpha_j(x)\big)\big] \in [0,1].$$

Definition 1.3 (Nerve complex) *Fix $\tau \in (0,1)$. The (weighted) nerve $\mathcal{N}_\tau$ has*

- *vertices $[m]$,*
- *an edge (i,j) when $w_{ij} > \tau$,*
- *a triangle (i,j,k) when all three of its edges are present.*

We truncate at 2-simplices ($\dim \mathcal{N}_\tau \le 2$).

During training we maintain the matrix $W = (w_{ij})$ and use it to build $\mathcal{N}_\tau$.

Enumerate vertices, edges, and triangles arbitrarily. Let $\mathcal{E}$ be the edge list and $\mathcal{F}$ the triangle list.

Definition 1.4 (Boundary operators)

$$B_1 \in \{-1, 0, 1\}^{|\mathcal{E}| \times m}, \qquad (B_1)_{(i,j),i} = -1,\ (B_1)_{(i,j),j} = +1;$$

$$B_2 \in \{-1, 0, 1\}^{|\mathcal{F}| \times |\mathcal{E}|},$$

where $(B_2)_{(i,j,k),(j,k)} = +1$, $(B_2)_{(i,j,k),(i,k)} = -1$, $(B_2)_{(i,j,k),(i,j)} = +1$ *for the orientation* $i < j < k$, *and* 0 *otherwise.*

Definition 1.5 (Hodge Laplacians)

$$\Delta_0 = B_1^\top B_1, \qquad \Delta_1 = B_1 B_1^\top \ + \ B_2^\top B_2.$$

The multiplicity of the zero eigenvalue of Δ_k *equals the* k*-th Betti number* b_k *[2, 1].*

References

[1] Georges De Rham. "Relations entre la topologie et la théorie des intégrales multiples". In: *L'Enseignement Mathématique* 35 (1936), pp. 213–228.

[2] Beno Eckmann. "Harmonische Funktionen und Randwertaufgaben in einem Komplex". In: *Commentarii Mathematici Helvetici* 17 (Dec. 1944), pp. 240–255.